Enhanced Construction Projects Compliance Prediction with Machine Learning

Adnan Abdelkarim Uzair Ahmad*

University of Doha for Science and Technology, Qatar

*Corresponding author: uzair.ahmad@udst.edu.qa.com

***Abstract*—This study investigates the use of machine learning to predict budget compliance in public construction projects using Portuguese procurement data. Building on prior work, we reproduce and enhance a Random Forest classifier by applying rigorous data preprocessing, SMOTE for class balancing, and systematic hyperparameter tuning. A refined classification target based on the Effective Price to Initial Price ratio allows the model to distinguish over-budget, compliant, and under-budget contracts. We further improve model interpretability through a multi-step feature selection process and unsupervised analysis, including UMAP and Kernel PCA. The resulting model achieves an accuracy of 89% and a weighted F1-score of 0.88—outperforming the original study. These results demonstrate that well-tuned classical models can deliver high performance while maintaining transparency, making them suitable for supporting early-stage decisions in public procurement.**

***Index Terms*—Machine Learning, Random Forest, Construction Projects, Budget Compliance, SMOTE, Dimensionality Reduction, Hierarchical Clustering, UMAP Projection, Kernel PCA.**

I. Introduction

Public construction projects often face challenges in maintaining budget compliance due to the complex nature of procurement, planning, and execution. These budget overruns not only affect project timelines and public perception but also lead to inefficient use of government resources. Identifying factors that contribute to cost deviations early in the procurement phase can help project stakeholders make informed decisions and potentially mitigate financial risks.

In recent years, machine learning (ML) has gained attention as a decision-support tool in the construction sector. By learning from historical project data, ML models can uncover patterns that might not be obvious through traditional statistical methods. Prior studies have demonstrated that such models can assist in predicting various project outcomes, including delays, cost overruns, and procurement risks.

A recent paper by Santos et al. (2024) explored the use of machine learning techniques to predict budget compliance in Portuguese public construction projects. Using the Portuguese Public Procurement Data (PPPData), the authors experimented with multiple classifiers and class balancing methods. The study reported that an Adam-based Artificial Neural Network (ANN) achieved the highest performance, with a weighted precision of 68.1%. While this result highlights the potential of ML in this context, the performance still leaves room for improvement—particularly when it comes to handling imbalanced datasets and selecting features with stronger predictive value.

By introducing a cleaner target classification, applying the Synthetic Minority Over-sampling Technique (SMOTE), and conducting class-overlap analysis, we aim to build a model that performs better and generalizes more reliably.

The motivation for these enhancements stems from several limitations observed in the original work. First, although multiple class balancing strategies were considered, their comparative impact on each model's predictive performance was not deeply analyzed. We selected SMOTE as our sole method to address class imbalance due to its effectiveness in generating synthetic examples for minority classes without removing valuable data from the majority class.

Second, the original feature selection process was largely based on correlation scores, which may not fully capture the nonlinear relationships relevant to a decision-tree-based model like Random Forest. We expanded this process by applying impurity-based importance (from the Random Forest itself) and permutation importance (a model-agnostic method) to identify and validate influential features from multiple perspectives.

Third, the original study did not include systematic hyperparameter tuning for the Random Forest classifier. Default parameters can often lead to suboptimal results, especially on imbalanced and noisy data. By applying GridSearchCV with a macro F1-score as the optimization metric, we aimed to improve generalization across all performance classes.

Finally, to better understand model behavior and the potential limitations of our features, we conducted a class-overlap analysis using statistical tests and visualizations. This helped assess the separability of the classes and provided insights into where the model might struggle.

Additionally, we explored the structure of the data using unsupervised learning and dimensionality reduction techniques. Hierarchical clustering, UMAP (Uniform Manifold Approximation and Projection), and Kernel PCA (Principal Component Analysis) were applied to assess class separability and latent structure. These exploratory steps offered further understanding of the data distribution, which informed model design and evaluation.

Our main contributions are as follows:

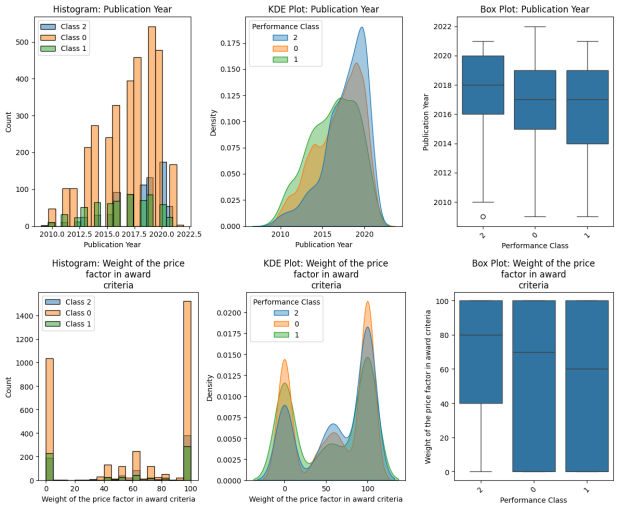
- We reproduce the results of the Random Forest classifier from the original study using the same PPPData data set.
- We introduce a refined target variable, "Performance Class," based on the Effective Price to Initial Price (EP/IP) ratio.
- We apply SMOTE to address class imbalance and conduct comprehensive feature selection using three complementary methods.
- We apply unsupervised clustering and dimensionality reduction methods to explore the internal structure of the dataset.
- We optimize the Random Forest model using GridSearchCV and evaluate its performance with detailed metrics and class-overlap analysis.

This work demonstrates that, with careful tuning and preprocessing, classical ensemble methods like Random Forest can outperform more complex architectures in structured data settings. The findings support the integration of ML tools into procurement workflows and highlight the need for data-driven policies in public project planning.

II. Exploratory Data Analysis

We used histograms, kernel density estimation (KDE) plots, and box plots to examine the distribution of feature values across the three classes. These plots were generated for each of the eight selected features.

- Histogram and KDE Plots: Provided a smoothed estimate of the feature distribution by class. Significant overlap was observed in several features, especially for Number of Bidders, where the distributions of all three classes were nearly indistinguishable
- Box Plots: Illustrated the spread and central tendency of feature values across classes. The interquartile ranges of many features overlapped substantially, suggesting limited class separation in parts of the feature space.



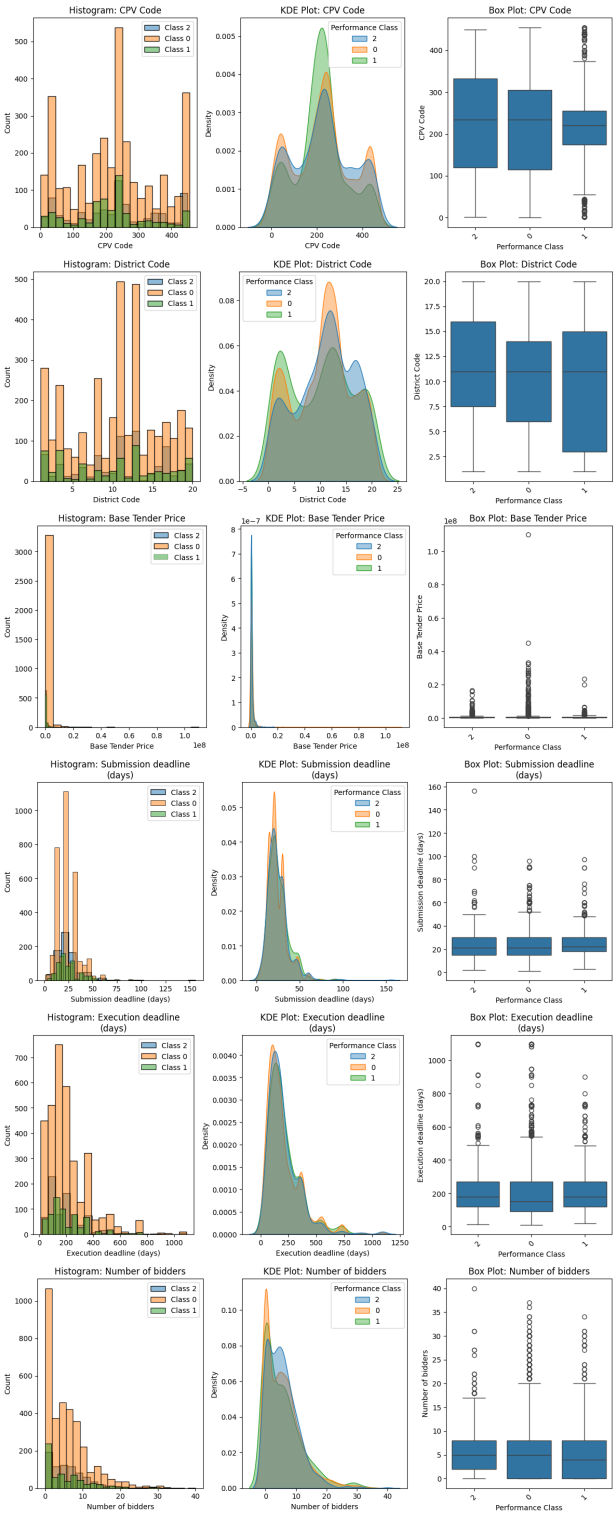


Fig. 1: Exploratory data analysis plots: Histogram, KDE, and Box plots for selected features.

III. Methodology

A. Dataset Overview

This study uses the Portuguese Public Procurement Data (PPPData), a structured dataset of construction contract records published by public entities. The dataset initially contained

5,214 contracts. Each record includes features related to tender value, deadlines, geographic location, award criteria, and contractor bids. The target of interest is a performance indicator derived from the ratio of the Effective Price (EP) to the Initial Price (IP).

B. Data Cleaning and Preprocessing

To ensure high-quality inputs for the machine learning model, we performed data cleaning by removing entries with missing or non-applicable values in four critical columns:

- Submission deadline (days)
- District Code
- Base Bid Price (€)
- Publication Year

This step excluded 441 records, reducing the dataset to 4,773 contracts. We then applied label encoding to convert categorical fields such as District Code and Award Criteria Class into numerical values compatible with the Random Forest algorithm.

TABLE I: Summary of Dataset Features

Column Name	Data Type	Description
Environmental criteria (T/F)	Integer	Indicates if the environmental criterion was considered (1 = Yes, 0 = No)
Publication Year	float64	Year the tender was published
District Code	float64	Code for the district (1–20)
Submission deadline (days)	float64	Days to submit applications (bids)
Execution deadline (days)	Integer	Days allowed to complete the project
Base Tender Price	float64	Maximum budget set by the client
Initial price (€)	float64	Initial agreed price
I. price Class	Integer	Based on initial price: 1: (<250k), 2: (250k–1M), 3: (>1M)
Award criteria class	Integer	Award type: 1 (Multi-factor), 2 (Lowest price), 0 (Missing)
Weight of price factor	float64	Weight of price factor (%)
Number of bidders	Integer	Number of submitted bids
CPV Code	Integer	Procurement code (Construction)
Performance Class (Target)	Integer	spill (≥105%), compliant (95%–105%), savings (≤95%)

C. Target Variable Construction

We introduced a refined classification target, Performance Class, derived from the EP/IP ratio. This metric captures deviations from the planned budget and allows us to categorize contract performance into three discrete classes:

TABLE II: Performance Class Coding Description

Description	Coding Scheme
The class given to the result of the project	Integer
Price spill: EP/IP $\geq$ 105%	
Price compliance: 95% $<$ EP/IP $<$ 105%	
Price savings: EP/IP $\leq$ 95%	

D. Addressing Class Imbalance

The class distribution was significantly imbalanced, with the "Spill" class dominating the dataset. Before model training, the target distribution was:

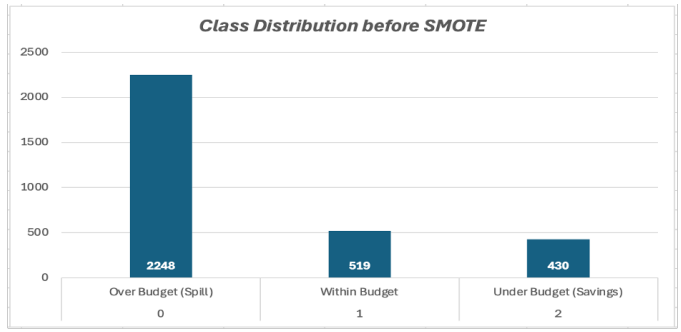


Fig. 2: Class distribution before SMOTE

To address this, we applied Synthetic Minority Over-sampling Technique (SMOTE) to the training data. SMOTE generates synthetic samples for minority classes based on feature-space similarity, improving model learning without discarding majority-class data.
After applying SMOTE, the class distribution was fully balanced:

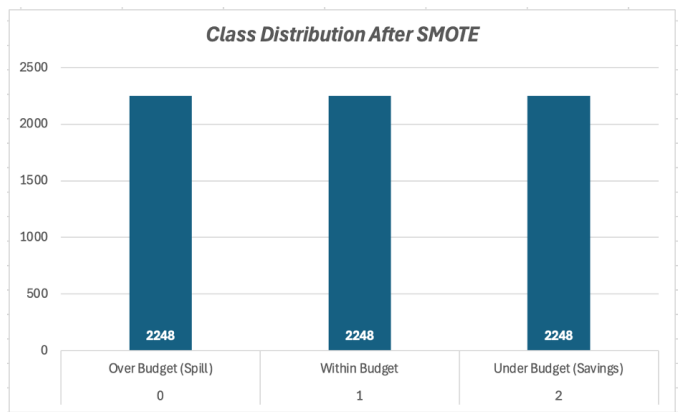


Fig. 3: Class distribution after SMOTE

E. Feature Selection

To improve model performance and reduce dimensionality, we used a multi-step feature selection process combining three methods:

- Correlation Heatmap
  A correlation matrix was used to examine linear relationships between features and the target variable. Although no feature had strong correlation, Publication Year (0.075) and Submission deadline (days) (0.052) showed modest association and were retained for further analysis.

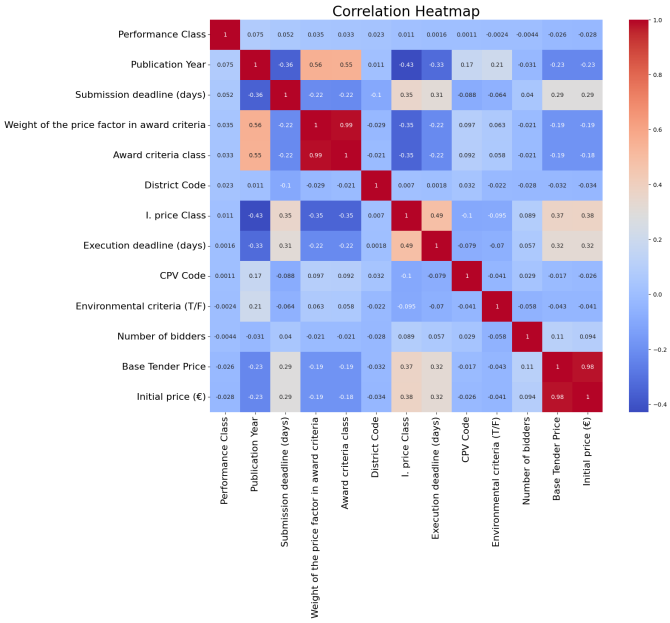


Fig. 4: Correlation heatmap showing relationships between features.

- Impurity-Based Feature Importance (Random Forest)
  We trained an initial Random Forest on the resampled data and ranked features by their contribution to reducing Gini impurity. Publication Year and Weight of the Price Factor emerged as the top predictors, contributing 32.4

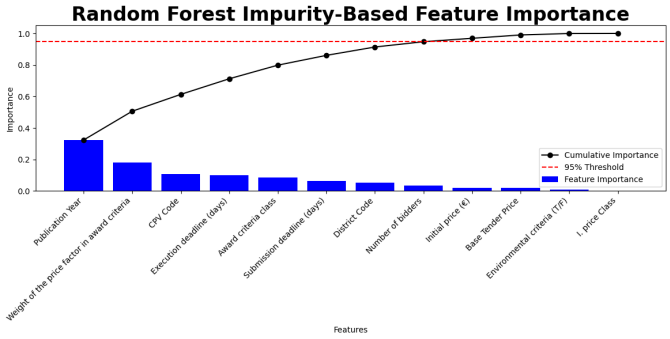


Fig. 5: Impurity-based feature importance rankings

- Permutation Importance (Random Forest)
  To validate the consistency of feature importance, we used permutation analysis on the test set. Results generally agreed with the impurity-based method, though differences appeared—highlighting features such as Award Criteria Class and Weight of the Price Factor as highly informative.

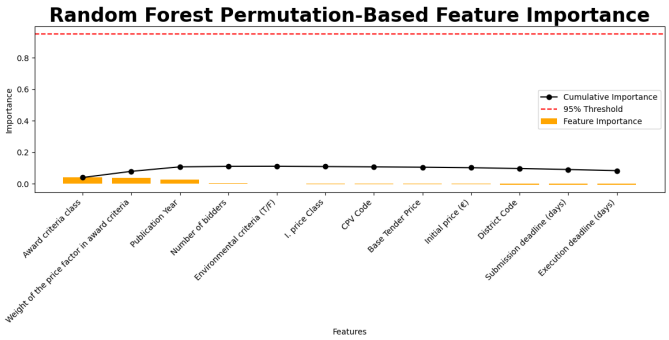


Fig. 6: Permutation-based feature importance rankings

Based on these analyses, the final eight features used for model training and testing were:

- Publication Year
- Weight of the Price Factor in Award Criteria
- District Code
- Base Tender Price
- Submission Deadline (days)
- Execution Deadline (days)
- Number of Bidders

These were applied consistently to both the resampled training data and the test set to ensure alignment in feature space.

F. Model Development and Hyperparameter Tuning

We selected the Random Forest Classifier for its ability to handle mixed data types, robustness to overfitting, and interpretability in feature ranking. To maximize performance, we performed hyperparameter tuning using GridSearchCV with 5-fold cross-validation and macro-averaged F1-score as the scoring metric. The parameter grid explored:

- `n_estimators = {100, 150, 200}`
- `max_depth = {None, 10, 20, 30}`
- `min_samples_split = {2, 5, 10}`
- `min_samples_leaf = {1, 2, 4}`
- `class_weight = {balanced, balanced_subsample}`
- `max_features = {sqrt, log2}`

The best-performing configuration was:

- n_estimators = 150
- max_depth = None
- min_samples_split = 2
- min_samples_leaf = 1
- class_weight = balanced_subsample
- max_features = sqrt

This setup was selected for its balance between accuracy, recall, and robustness to class imbalance, particularly when evaluating minority classes.

G. Unsupervised Analysis and Dimensionality Reduction

To explore the latent structure of the data and validate class separability beyond supervised modeling, we applied a combination of clustering and dimensionality reduction

techniques.

Hierarchical Clustering: We performed agglomerative hierarchical clustering on the dataset using Ward's method, which minimizes within-cluster variance. A dendrogram was generated to visualize the hierarchical relationships among data points based on the selected features.

UMAP Projections: Uniform Manifold Approximation and Projection (UMAP) was applied to reduce the high-dimensional feature space into two dimensions. To test robustness, this analysis was conducted both before and after applying RobustScaler to the data. UMAP projections were visualized using scatter plots with color-coded points based on the "Performance Class" to assess how well the classes are grouped.

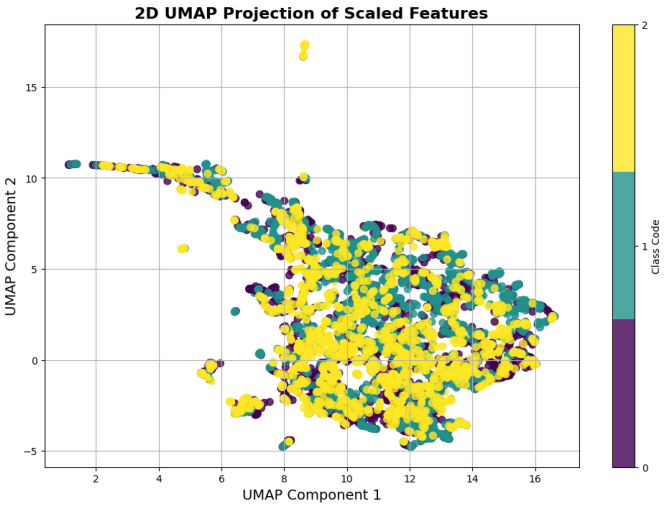


Fig. 7: UMAP projection of scaled features

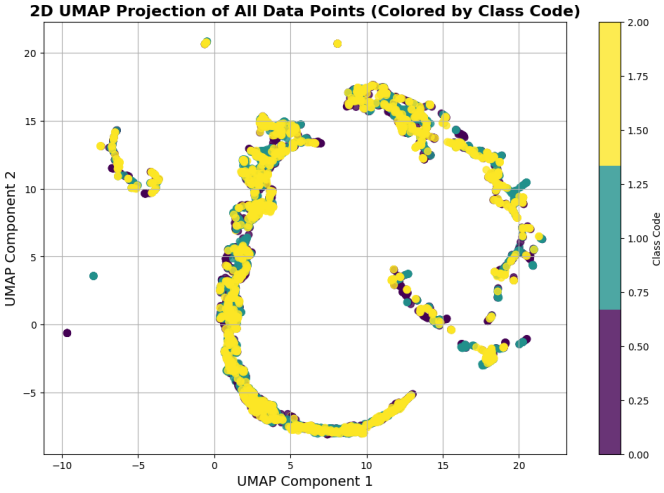


Fig. 8: UMAP projection of all data points

Kernel PCA: Additionally, Kernel Principal Component Analysis (PCA) with an RBF kernel was used to map the data into a lower-dimensional space. The two-dimensional transformation was visualized using scatter plots, helping to reveal nonlinear structures and potential class boundaries.

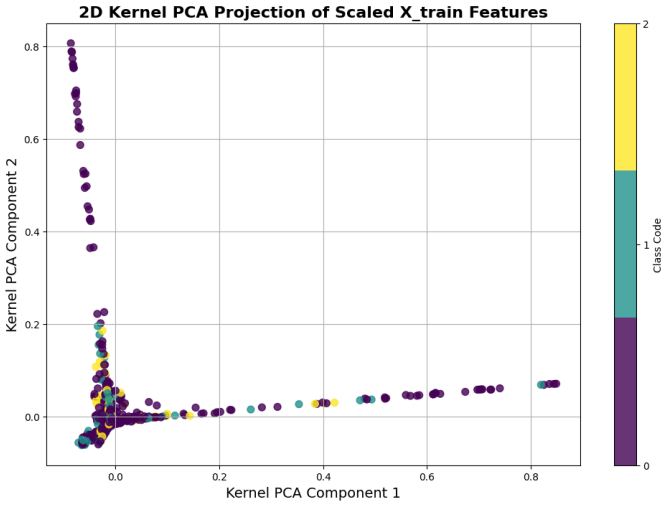


Fig. 9: Kernel PCA projection

These visual analyses provided insights into data distribution and class separability, helping to validate observations from the class-overlap analysis and inform model design choices.

TABLE III: Classification Performance Metrics

Class	Precision	Recall	F1-Score	Support
0 (Over Budget)	0.90	0.95	0.93	1108
1 (Within Budget)	0.83	0.71	0.77	219
2 (Under Budget)	0.84	0.76	0.80	248
Accuracy	–	–	0.89	1575
Macro Avg	0.86	0.81	0.83	1575
Weighted Avg	0.88	0.89	0.88	1575

IV. Results and Evaluation

This section presents the performance of the Random Forest classifier trained on the resampled dataset using the selected features and tuned hyperparameters. Evaluation was conducted on the held-out test set, using standard classification metrics to assess the model's predictive performance.

A. *Classification Performance*

The trained model achieved the following metrics on the test set:

TABLE IV: Model Evaluation Metrics

Metric	Value
Test Accuracy	0.89
Precision (Weighted)	0.88
Recall (Weighted)	0.89
F1-Score (Weighted)	0.88

These results indicate that the model performs well in identifying all three performance classes, maintaining a

balance between precision and recall. The high F1-score suggests that the model is not only precise in its predictions but also consistently able to detect relevant instances, particularly in the previously underrepresented classes.

The confusion matrix below shows the distribution of predicted versus actual labels for the three classes: Spill, Compliance, and Savings. This helps visualize the types of errors the model tends to make and whether misclassifications occur more frequently between certain pairs of classes.

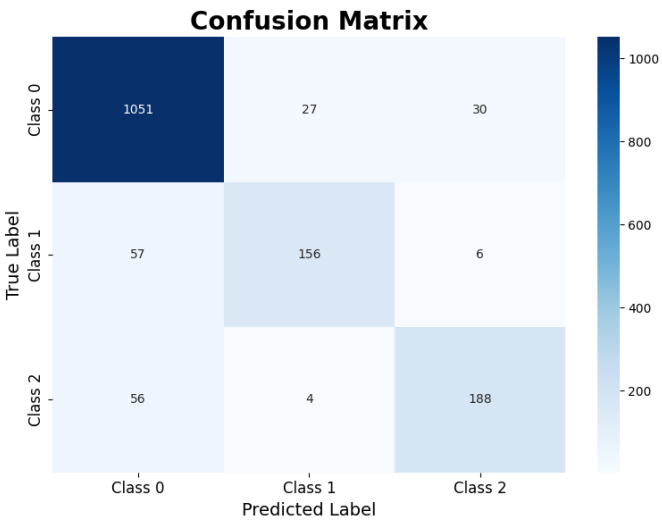


Fig. 10: Confusion matrix

In preliminary tests, the majority of predictions for the dominant class (Spill) were correctly classified. Most errors were observed between Compliance and Savings, which is consistent with class-overlap analysis discussed later.

B. Feature Importance Rankings

The following features were identified as the most important for classification based on impurity reduction and permutation importance:

1) **Publication Year**
   Captures regulatory or market changes over time that affect procurement outcomes.
2) **Weight of the Price Factor in Award Criteria**
   Indicates the role of cost competitiveness in winning bids.
3) **CPV Code**
   Reflects the category of work and may influence pricing variability and contract behavior.
4) **District Code**
   Accounts for regional effects in contract execution.
5) **Base Tender Price**
   Serves as a proxy for project scale.
6) **Submission Deadline (days)**
   May correlate with preparation quality and bid pricing.
7) **Execution Deadline (days)**
   Longer deadlines may introduce cost uncertainty.
8) **Number of Bidders**
   Indicates competition level, though its predictive power was limited by class overlap.

These rankings align with domain knowledge, confirming that temporal, regional, and financial variables play key roles in predicting budget performance.

C. Observations

- The Random Forest model performed well across all classes after applying SMOTE, supporting its effectiveness in handling imbalanced data.
- Precision and recall values remained closely matched, indicating the model did not favor any single class disproportionately.
- The classifier demonstrated good generalization, as evidenced by stable results across training and testing.

V. Class Overlap Analysis

To evaluate the separability of the three performance classes Spill, Compliance, and Savings, we conducted a class-overlap analysis using both visual and statistical techniques. This analysis helps determine how well the selected features distinguish between categories, and where the model may encounter ambiguity during classification.

TABLE V: Class Overlap Analysis Results

Test Type	Class Pair	Metric(s)
Bhattacharyya Coefficient	2 vs. 0	0.9839
	2 vs. 1	0.9575
	0 vs. 1	0.9886
Kolmogorov–Smirnov Test	2 vs. 0	D = 0.0808, p = 0.0005
	0 vs. 1	D = 0.0524, p = 0.1001
ANOVA Test	All Classes	F = 0.0480, p = 0.9531

A. Bhattacharyya Coefficient

To quantify the visual observations, we calculated the Bhattacharyya coefficient for each pair of classes. This coefficient measures the amount of overlap between two probability distributions, with values close to 1 indicating high similarity. For the feature Number of Bidders, the coefficients were:

- Spill vs. Savings: 0.9839
- Spill vs. Compliance: 0.9886
- Compliance vs. Savings: 0.9575

These values confirm that Number of Bidders offers limited discriminatory power due to the high degree of overlap across all class pairs.

B. Kolmogorov-Smirnov (KS) Test

The KS test was used to statistically assess the difference between cumulative distributions for each pair of classes. The null hypothesis is that the samples are drawn from the same distribution. For Number of Bidders:

- Spill vs. Savings: D = 0.0808, p = 0.0005 $\rightarrow$ statistically significant difference
- Spill vs. Compliance: D = 0.0524, p = 0.1001 $\rightarrow$ not statistically significant

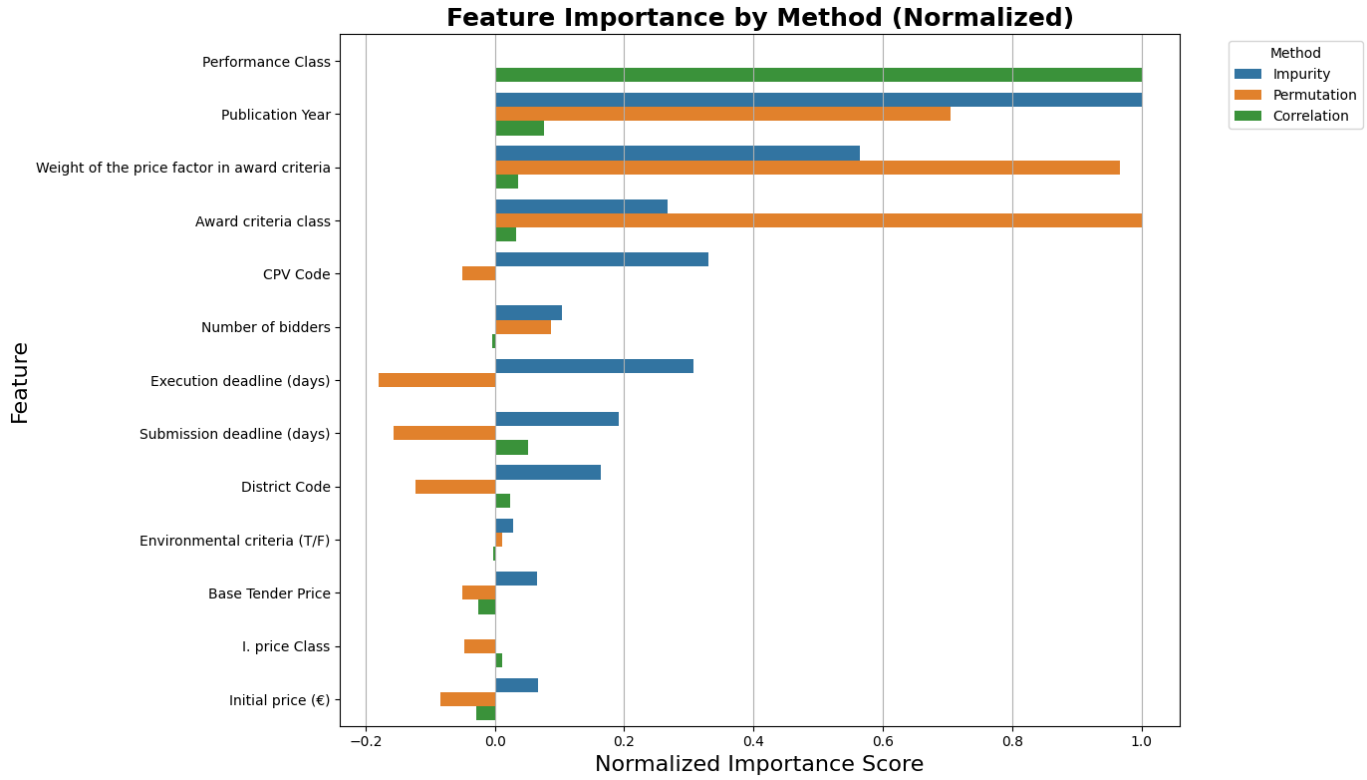


Fig. 11: Feature importance scores based on impurity reduction, permutation importance, and feature correlation analysis.

- Compliance vs. Savings: Similar non-significant differences observed

These results suggest that although small differences exist, they are not consistent or strong enough to support clear class separation based on this feature.

C. ANOVA Test

We conducted a one-way Analysis of Variance (ANOVA) to test for significant differences in mean values across the three classes. For the feature Number of Bidders, the test produced:

- F-statistic: 0.0480
- p-value: 0.9531

This p-value indicates no statistically significant difference in the means of this feature across the three classes, confirming that it does not contribute meaningfully to class discrimination in this context.

D. Implications

This overlap analysis reinforces that certain features—while relevant to the domain—may not effectively differentiate between performance outcomes on their own. In particular, Number of Bidders, despite its intuitive appeal as a proxy for competitiveness, shows minimal predictive value when used in isolation.

These findings support the use of ensemble models like Random Forest, which can leverage subtle feature interactions that may not be evident in univariate analysis. However, they also suggest that future work may benefit from engineering new features or incorporating contextual variables to improve class separation.

VI. Discussion

The results of this study demonstrate that a well-tuned Random Forest classifier, combined with appropriate preprocessing and feature selection, can outperform more complex models such as artificial neural networks in predicting construction contract budget compliance. By applying SMOTE to address class imbalance and refining the set of predictive features, we achieved a weighted F1-score of 0.88 and an accuracy of 0.89—both improvements over the baseline metrics reported in the original study.

One key insight is that model performance benefited significantly from the inclusion of multiple feature importance techniques during selection. While the original paper relied primarily on correlation analysis, our combination of impurity-based importance and permutation-based ranking allowed us to identify stable and informative predictors, such as Publication Year, Weight of the Price Factor, and CPV Code. This resulted in a more focused model with reduced noise and better generalization.

The application of SMOTE proved effective in correcting the imbalance among the three classes—Spill, Compliance, and Savings. Without this step, the model tended to favor the dominant class (Spill), reducing its utility in real-world scenarios where identifying minority classes is equally important. After balancing, the model maintained strong recall and precision

across all classes, suggesting a more equitable performance.

Our analysis also highlighted the limits of individual features in isolating class boundaries. Class overlap analysis showed that features like Number of Bidders exhibited nearly identical distributions across performance categories, limiting their utility in isolation. This reinforces the importance of using ensemble methods that can capture complex interactions between features, rather than relying on individual variables.

In addition to supervised modeling, we explored the data's latent structure through dimensionality reduction and clustering. UMAP projections—before and after scaling—helped visualize how data points relate in a lower-dimensional space and showed moderate clustering by class. Kernel PCA added further depth by highlighting potential nonlinear patterns. These visualizations helped validate our modeling choices and supported the hypothesis that class separability exists, though it may be subtle and distributed across multiple features. While the model performed well, there are several limitations to consider. First, the reliance on historical procurement data restricts generalizability to other regions or sectors without similar data characteristics. Second, some variables in the dataset may be influenced by external or unobserved factors (e.g., political or market conditions), which are not captured in the model. Finally, while we improved interpretability compared to neural networks, ensemble methods like Random Forest still lack full transparency, especially in feature interactions.

Despite these limitations, the study reinforces the practical value of machine learning in public procurement. A model with high accuracy and balanced performance can support early-stage decisions by flagging projects likely to deviate from budget targets. This has direct implications for policy design, risk management, and contract evaluation. Future research may explore more interpretable models, such as rule-based classifiers or SHAP-based analysis for feature explanations. Additionally, incorporating textual procurement documents or expanding the feature set with external economic indicators could improve predictive power.

Compared to the original study, our approach yields notable improvements in both predictive performance and model transparency. While the original authors reported a validation accuracy of 83.2% and an F1-score of 0.846 using a Random Forest classifier, our tuned model achieved 89% accuracy and an F1-score of 0.88. This gain is attributed to the use of deeper trees, more estimators (150 vs. 33), and systematic hyperparameter tuning via GridSearchCV with a broader search space. In contrast, their Random Forest configuration was restricted to shallow trees (max depth = 4) and fewer estimators, potentially limiting the model's ability to learn complex patterns.

Beyond model performance, our work introduces several additional analytical components that were not present in the original study. These include class-overlap analysis using statistical measures (Bhattacharyya Coefficient, Kolmogorov–Smirnov Test, and ANOVA), and unsupervised learning techniques (hierarchical clustering, UMAP, and Kernel PCA) to visualize the latent structure of the data. These methods provided valuable insights into the feature space and class distribution, strengthening the rationale behind our modeling decisions.

While both studies applied SMOTE for class balancing and similar feature selection techniques (correlation, impurity-based, and permutation importance), our extended analysis and visualization pipeline enhanced interpretability and model design. The decision to use Random Forest as the final model, rather than a more complex neural network, also aligns with our goal of balancing performance with explainability—critical in public sector applications where model transparency is essential.

VII. Conclusion

This study reproduced and extended prior research on predicting budget compliance in public construction contracts using Portuguese procurement data. By focusing on a Random Forest classifier, we demonstrated that classical ensemble models—when properly tuned—can outperform more complex architectures in structured data settings. Our model achieved 89Key improvements came from a well-defined target variable, comprehensive feature selection, systematic hyperparameter tuning, and targeted class balancing using SMOTE. In addition, we introduced exploratory analysis techniques—such as UMAP, Kernel PCA, and hierarchical clustering—to visualize the data's latent structure and assess class separability. Statistical tests further validated areas of class overlap and informed the modeling process. While limitations remain, particularly in generalizing to other datasets or contexts, the findings support the use of interpretable, high-performing models in public procurement workflows. This has practical implications for early-stage contract evaluation and risk management. Future work may focus on improving model interpretability using post-hoc explanation tools, extending the feature space with external variables, or applying similar techniques to other domains where structured procurement data is available.

References

[1] Batista, G. E. a. P. A., Prati, R. C., & Monard, M. C. (2004). A study of the behavior of several methods for balancing machine learning training data. ACM SIGKDD Explorations Newsletter, 6(1), 20–29. https://doi.org/10.1145/1007730.1007735

[2] Bergstra, J., & Bengio, Y. (2012). Random search for Hyper-Parameter Optimization [Journal-article]. Journal of Machine Learning Research, 13, 281–305. https://www.jmlr.org/papers/volume13/bergstra12a/bergstra12a.pdf

[3] Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.

[4] Buda, M., Maki, A., & Mazurowski, M. A. (2018). A Systematic Study of the Class Imbalance Problem in Convolutional Neural Networks. Neural Networks, 106, 249–259.

[5] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research, 16, 321–357.

[6] De Sousa, L. J., Martins, J. P., & Sanhudo, L. (2024). Predicting construction project compliance with machine learning model: case study using Portuguese procurement data. Engineering Construction & Architectural Management, 31(13), 285–302. https://doi.org/10.1108/ecam-09-2023-0973

[7] Diaz-Papkovich, A., Anderson-Trocme, L., Ben-Eghan, C., & Gravel, S. (2019). UMAP reveals cryptic population structure and phenotype heterogeneity in large genomic cohorts. PLoS Genetics, 15(11), e1008432. https://doi.org/10.1371/journal.pgen.1008432

[8] Drummond, C., Institute for Information Technology, National Research Council Canada, Holte, R. C., & Department of Computing Science, University of Alberta. (2003). C4.5, Class Imbalance, and Cost Sensitivity: Why Under-Sampling beats Over-Sampling. In Workshop on Learning From Imbalanced Datasets II, ICML, Washington DC, 2003. https://www.site.uottawa.ca/ nat/Workshop2003/drummondc.pdf

[9] Douzas, G., Bacao, F., & Last, F. (2018). Improving imbalanced learning through a heuristic oversampling method based on k-means and SMOTE. Information Sciences, 465, 1–20. https://doi.org/10.1016/j.ins.2018.06.056

[10] Fawcett, T. & Silicon Valley Data Science. (2006). An introduction to ROC analysis. In Pattern Recognition Letters [Journal-article]. https://doi.org/10.1016/j.patrec.2005.10.010

[11] Guyon, I., & Elisseeff, A. (2003). An Introduction to Variable and Feature Selection. Journal of Machine Learning Research, 3, 1157–1182.

[12] Jain, T., Misra, S., & Truong, D. (2022). Using deep learning to predict unstable approaches for general aviation aircraft. Journal of Aerospace Information Systems, 19(12), 811–817. https://doi.org/10.2514/1.i011132

[13] King, G., & Zeng, L. (2001). Logistic regression in rare events data. Political Analysis, 9(2), 137–163. https://doi.org/10.1093/oxfordjournals.pan.a004868

[14] McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. arXiv preprint arXiv:1802.03426.

[15] Mika, S., Scholkopf, B., Smola, A., Muller, K.-R., Scholz, M., Riitsch, G., & GMD FIRST. (2001). Kernel PEA and De-Noising in feature spaces. GMD FIRST. https://papers.nips.cc/paper_files/paper/1998/file/226d1f15ecd35f784d2a20c3ecf56d7f-Paper.pdf

[16] Molnar, C. (n.d.). Interpretable Machine Learning. https://christophm.github.io/interpretable-ml-book/

[17] Ribeiro, M. T., et al. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining - KDD '16, pp. 1135–1144. https://doi.org/10.1145/2939672.2939778

[18] Schölkopf, B., Smola, A., & Müller, K.-R. (1998). Nonlinear component analysis as a kernel Eigenvalue problem. Neural Computation, 10, 1299–1319. https://www.mlpack.org/papers/kpca.pdf

[19] Strobl, C., Boulesteix, A.-L., Kneib, T., Augustin, T., & Zeileis, A. (2008). Conditional variable importance for random forests [Methodology article]. BMC Bioinformatics. https://doi.org/10.1186/1471-2105-9-307

[20] The analysis of variance. (n.d.). Google Books. https://books.google.com.qa/books?id=z9yUEAAAQBAJ

[21] The annals of Mathematical Statistics (By Institute of Mathematical Statistics & JSTOR). (n.d.). https://www.jstor.org

[22] Tipping, Michael E., & Christopher M. Bishop. (1999). Probabilistic Principal Component Analysis. Journal of the Royal Statistical Society: Series B (Statistical Methodology), vol. 61, no. 3, pp. 611–622. https://doi.org/10.1111/1467-9868.00196.